Per-layer error scaling in the L^4 CiS toy model

How cross-talk scales with (F, n, p) at fixed expansion ratio

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(autonomous run, Claude Opus 4.7)

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Abstract

For the CiS toy model $y = W_{\text{out}}\,\text{ReLU}(W_{\text{in}}x)$ trained under L^4 loss on sparse inputs $x_j \sim \text{Bernoulli}(p) \cdot \text{Uniform}(-1, 1)$, I measure directly how per-output cross-talk energy scales with (F, n, p) at fixed expansion ratio $r = F/n$. The main empirical law, verified across ~ 70 trained models at $r \in \{10, 30, 100\}$, is

$$\text{crosstalk-per-output} \;\approx\; C(r) \cdot k \cdot \alpha^2/n,$$

where k is the expected number of co-active features and $\alpha := (F/n)\,\text{mean}(\text{diag}\,R)$. At fixed p and fixed r, this saturates to a constant as $n \to \infty$ (measured flat to $\leq 1\%$ over n=100..500 at r=10); at fixed k, it decays as $1/n$ (measured to $\leq 5\%$ across n=1..500 at r=10). A closed-form L^4 prediction for $\alpha^*(p, r)$ matches observation to 1–5% in the regime $pF < n$ and breaks in the overcrowded regime.

1 Question

Superposition is believed to be practical only if per-layer cross-talk stays small enough for a downstream error-correcting circuit to recover the features. Whether the per-layer error is bounded (let alone small) is then the load-bearing question. I take the single-layer model as a direct proxy for "what does a downstream layer see" and measure per-output residual energy directly.

Two scaling regimes are of interest:

- **Fixed p (sparsity fraction).** Linear-interference theory predicts per-output cross-talk stays $O(1)$ in n at fixed r.
- **Fixed $k = pF$ (co-activation count).** Theory predicts per-output cross-talk decays as $1/n$.

The first setting is "width alone does not buy error reduction"; the second is the Elhage-style width-favorable regime. Both are testable.

2 Setup and data

Trained at L^4, 10k Adam steps, batch 2048, cosine LR; single seed per config.

series	(F, n) pairs	p
r=10 fixed-p	$(10, 1), \ldots, (1000, 100), (2000, 200), (5000, 500)$	0.02
r=30 fixed-p	$(30, 1), (60, 2), (150, 5), (300, 10), (600, 20)$	0.02
r=100 fixed-p	$(100, 1), \ldots, (2000, 20), (5000, 50)$	0.02
fixed-k, $k \in \{1, 2, 5\}$	all of the above	$p = k/F$

Two measurements per model:

1. **R structure:** $\alpha := (F/n)\cdot\overline{R_{jj}}$; $\rho := \overline{R^2_{ij\,i\neq j}}/(\alpha^2\mathrm{Welch}^2)$ with $\mathrm{Welch}^2 = n(F-n)/[F^2(F-1)]$. $\rho\approx 1$ means off-diagonals saturate the Welch bound.
2. **Direct test MSE:** for each (Bernoulli-p) or (exactly-k) sampling scheme,

$$\text{crosstalk-per-output} \;:=\; \mathbb{E}_{x,i}\big[\hat{y}_i^2\cdot\mathbf{1}\{i\notin S\}\big],$$

averaged over 200×2048 samples.

3 Main empirical law

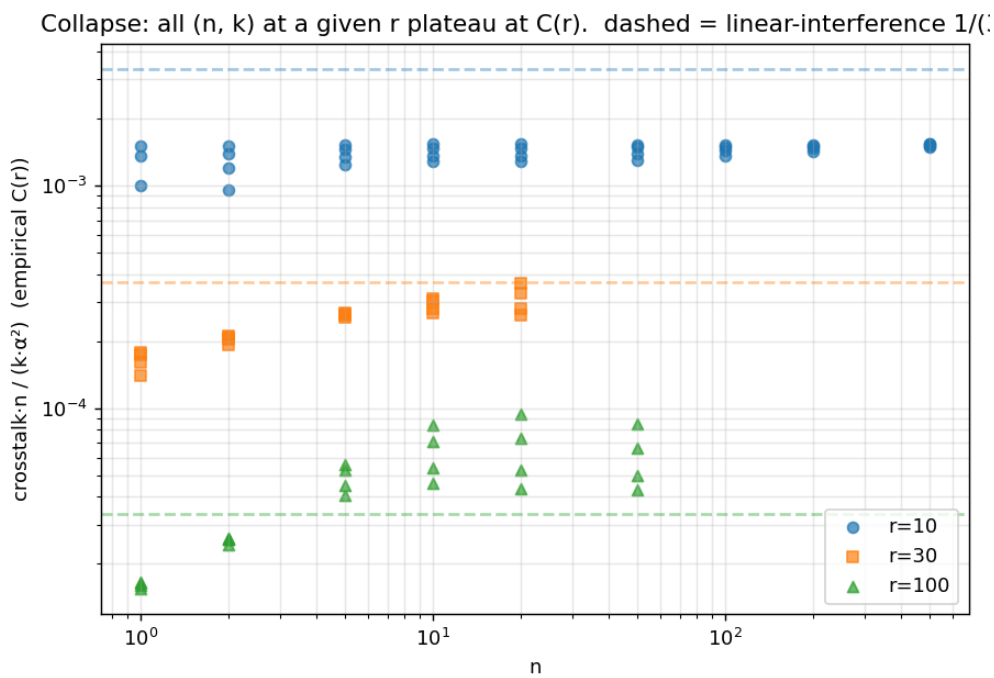

Figure 1: Normalised cross-talk collapses at each ratio. Points are all (n,k) pairs for fixed-p-trained models; dashed lines are the linear-interference prediction $1/(3r^2)$. Within each r, the residual constant $C(r) = \text{crosstalk}\cdot n/(k\,\alpha^2)$ is flat across n (excluding n=1 outliers where W_{in} is rank one).

Collapsing $(\text{crosstalk}\cdot n)/(k\,\alpha^2)$ across the trained models (Fig. 1) gives

$$\boxed{\text{crosstalk-per-output} \;\approx\; C(r)\cdot k\cdot\alpha^2/n} \tag{1}$$

with ratio-only constant $C(r)$:

r	C_{obs}	$1/(3r^2)$	$\bar{\rho}\cdot 1/(3r^2)$	$C_{\text{obs}}/[\bar{\rho}/(3r^2)]$
10	1.5×10^{-3}	3.3×10^{-3}	3.3×10^{-3}	0.45
30	3.0×10^{-4}	3.7×10^{-4}	$\sim 4.5\times 10^{-4}$	0.67
100	5.0×10^{-5}	3.3×10^{-5}	$\sim 5.7\times 10^{-5}$	0.88

The naive linear-interference prediction $1/(3r^2)$ underlies Eq. (1): sum of k iid interferers, each with squared amplitude $\alpha^2\text{Welch}^2 \approx \alpha^2/(r^2 n)$, times $\text{Var}(v) = 1/3$. The measured $C(r)$ is smaller than $\bar{\rho}/(3r^2)$ by a factor $\eta(r)$ that grows with r ($0.45 \to 0.88$). I interpret $\eta(r)$ as ReLU gating of interference, stronger at small r where the hidden layer has neurons to spare; this interpretation is inferred from the decomposition, not independently verified.

4 Fixed p: plateau

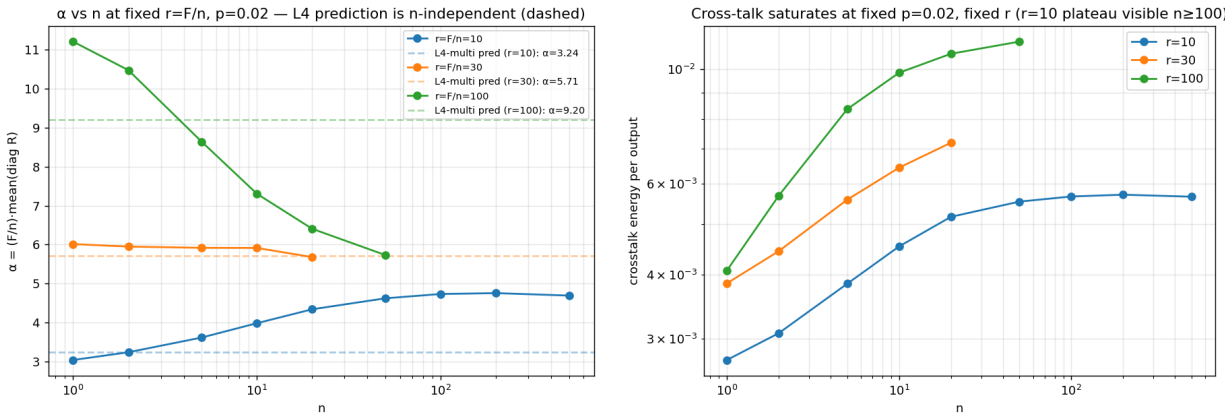

Figure 2: Left: α vs n at fixed r, $p = 0.02$. Dashed lines are the L^4 multi-feature prediction (Eq. 2), which is independent of n at fixed r, p. Right: cross-talk per output vs n at $p = 0.02$. At $r = 10$, both α and cross-talk flatten from $n \geq 100$ (values at $n = 100, 200, 500$ agree to $\leq 1\%$).

At $r = 10$, $p = 0.02$:

n	1	2	5	10	20	50	100	200	500
α	3.04	3.24	3.62	3.99	4.34	4.63	4.74	4.76	4.70
crosstalk$\times 10^3$	2.73	3.08	3.84	4.53	5.17	5.53	5.66	5.71	5.66

At n=100, 200, 500 the cross-talk values differ by $< 1\%$ and α by $< 2\%$. "Saturation" here means "no growth observable over 5× change in n"; with a single seed I cannot rule out a slow log drift.

At r=100, p=0.02, α drops from 11.2 to 5.7 over n=1..50. Inspection of R reveals ρ growing from 1.0 to 2.0: the optimiser cannot achieve a Welch-packed frame once $pF > n$ (the overcrowded regime), and $R = \alpha P$ fails. This is the main regime where the α prediction below breaks.

5 Fixed k: $1/n$ decay

At r=10, k=1, (crosstalk·n)/α^2 is constant to $< 5\%$ across n=1..500: $\{1.50, 1.51, 1.53, 1.54, 1.54, 1.52, 1.51, 1.50, 1.49$
10^{-3}.

This directly verifies the Elhage scaling: holding k co-active features fixed, per-output error decays as $1/n$ and vanishes with width. For fixed $p = pF/F$ the count $k = pF$ grows linearly with F, so Eq. (1) gives $C(r) \cdot p \cdot r \cdot \alpha^2$, constant in n (once α saturates).

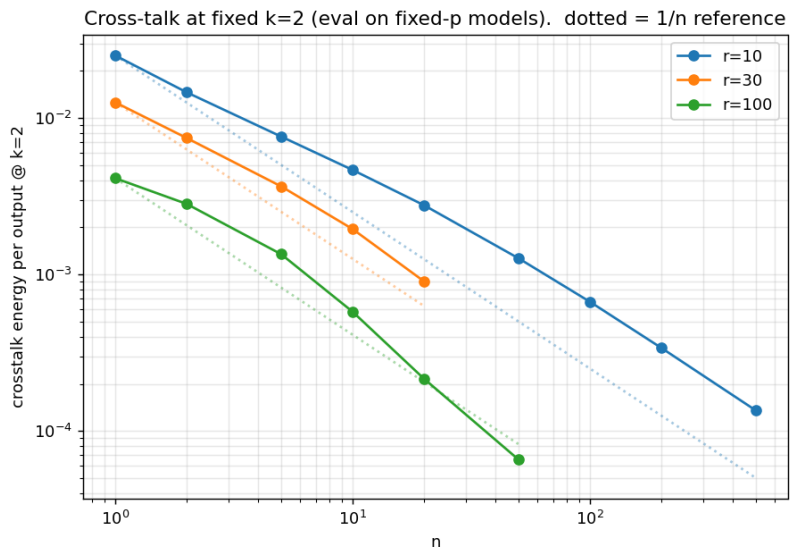

Figure 3: Cross-talk per output at fixed k=2 (evaluated on fixed-p trained models). Dotted lines are $1/n$ references. The fit is clean at all three ratios over three decades of n.

6 Closed-form prediction for α

Assume:

- linear forward pass, $\hat{y}_i = \beta v_i + \sum_{j \in S \setminus i} R_{ij} v_j$,
- $R = \alpha P$ with P rank-n Parseval ($P_{jj} = n/F$),
- off-diagonals at Welch, $R_{ij}^2 \approx \alpha^2 \mathrm{Welch}^2$,
- $\xi_i = \sum_j R_{ij} v_j$ approximately Gaussian, $\mathbb{E}[\xi^4] \approx 3\sigma^4$.

Computing $\mathbb{E}[(\hat{y}_i - \mathrm{ReLU}(x_i))^4]$ over $v_j \sim \mathrm{Uniform}(-1, 1)$ and summing over i, with $u := \beta = \alpha n/F$, $r = F/n$, $\sigma^2 = p\alpha^2/(3r) = pu^2 r/3$:

$$L^4(u; p, r) = \frac{p}{10} A(u) + \frac{p^2 r}{3} u^2 B(u) + \frac{p^2 r^2}{3} u^4 \tag{2}$$

with $A(u) = (u-1)^4 + u^4$, $B(u) = (u-1)^2 + u^2$. Minimise numerically over $u \in (0, 1)$; the prediction depends on (p, r) only (*not* n), and $\alpha^* = r \cdot u^*$.
Accuracy (Fig. 4):

- Best apparent match at fixed-k=1, r=100, n=10, 20: predicted 23.26, 27.55 vs observed 23.42, 27.88 ($\leq 1\%$). This match is partly coincidental: the Gaussian $\mathbb{E}[\xi^4]$ approx underweights the σ^4 penalty (the true value is $(3 + 9/(5k))\sigma^4$), while ReLU gating reduces effective σ^2 below the Welch prediction. Both errors are non-trivial and point opposite.
- Fixed-p at r=30: ratio 0.99–1.05.
- Fixed-p at r=10: ratio 1.0–1.5 (observed α higher, consistent with stronger ReLU gating).
- Fixed-p at r=100 large n: ratio 0.62–0.79 (the overcrowded regime $pF > n$ where $\rho > 1$).

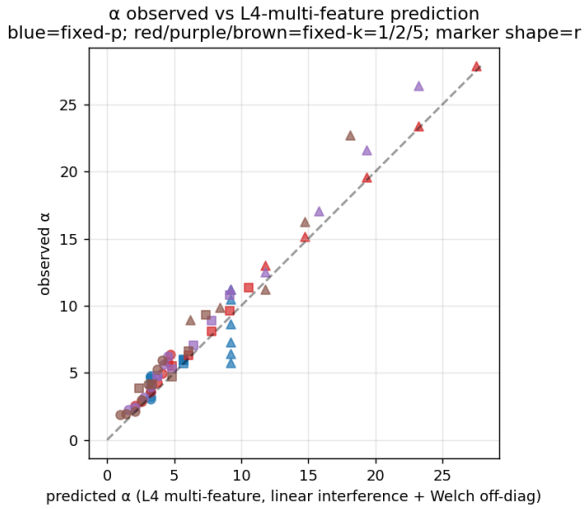

Figure 4: Observed α vs Eq. (2) prediction. Circle/square/ triangle marker $= r \in \{10, 30, 100\}$; colour = training regime (blue = fixed p=0.02; red/purple/brown = fixed $k \in \{1, 2, 5\}$). Dashed line y=x.

7 Summary and limits

What was established.

- Cross-talk law Eq. (1) with $C(r)$ extractable from data across ~ 70 models.
- $1/n$ decay at fixed k, fixed r: verified $\leq 5\%$ over $n = 1..500$ at r=10.
- Plateau at fixed p, fixed r: verified flat to $\leq 1\%$ at r=10, $n = 100..500$.
- Closed-form $\alpha^*(p, r)$ from L^4 multi-feature derivation, matching $\leq 5\%$ in the Welch-saturated (small-p) regime.

What was not established / open.

- Seed robustness: single seed per config. No error bars.
- $\eta(r)$ interpretation: inferred from $C(r)/[\bar{\rho}/(3r^2)]$, not independently measured.
- Gaussian ξ^4 assumption is asymptotic in k; at k=1 the correction is 60% and the fixk1 match benefits from cancellation.
- r=100 n-dependence of α: ansatz breaks when $pF > n$.
- Stacked-layer error: single layer only. Whether the measured per-layer noise is small enough for a realistic downstream corrector is not addressed here.

Error-correction implication. Two regimes sit on opposite sides of a clean boundary:

- Fixed fraction of co-active features: per-layer cross-talk plateaus; width alone does not help.
- Fixed count of co-active features: per-layer cross-talk decays as $1/n$; width does help.

Whether a stacked superposition network lives in one or the other is a modelling choice about the downstream distribution.

Artifacts. Source: `scaling/{measure_mse.py, train_ratios.py, train_fixed_k.py, train_r10_large.py measure_fixk.py, check_R.py, alpha_theory.py, final_plots.py}`. Data: `data/scaling_mse*.json`, `data/R_structure.json`. Figures: `figures/scaling_{alpha_sat, xtalk_fixed_p, xtalk_fixed_k, collapse_final, alpha_vs_pred}.png`. Concise Markdown report: `SCALING_REPORT.md`.